MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FIRST

SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCZGY101 – Biological Basis of Behaviour 1

Duration: 1.½ hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks Questions

Answer **All** Questions

Each question carries **1** mark

1. The secretions of endocrine glands are called------ [Remember] [CO: 1]
2. The first hominin to emigrate out of Africa was------ [Remember] [CO: 1]
3. The deeply stained angular bodies present in the cytoplasm of neuron are ------ [Remember] [CO: 2]
4. The Goldman equation is used to calculate the -------- of a cell's membrane. [Remember] [CO: 2]
5. Parkinson's disease is due to degeneration of neurons that release the neurotransmitter --------- [Remember] [CO: 2]
6. The -------- is a canal in the midbrain connecting the third and fourth ventricles. [Understand] [CO: 4]
7. The -------- is the main control center for the autonomic nervous system located in the brain. [Remember] [CO: 3]
8. The ------- horn of the spinal cord gray matter is involved in processing sensory information. [Remember] [CO: 3]
9. There are ------ pairs of spinal nerves. [Remember] [CO: 4]
10. The ------ nerve is responsible for controlling the muscles of the larynx. [Remember] [CO: 3]

[1 x 10 = 10]

Part **B**

Short Answer Type Questions

Answer **10** Questions

Each question carries **2** marks

11. How diet quality relate to the development of large size brain? [Understand] [CO: 1]
12. How can we ensure the well-being of animals used in research? [Understand] [CO: 1]
13. Explain the scope of neuroscience in research fields. [Understand] [CO: 1]
14. Comment on the types of neuroglial cells. [Understand] [CO: 2]
15. How do sensory neurons contribute to the functioning of the nervous system? [Understand] [CO: 2]
16. How do changes in external potassium concentration affect resting potential? [Understand] [CO: 2]
17. What factors influence the speed of action potential propagation? [Understand] [CO: 2]

18. What is the role of NMDA receptors in synaptic plasticity? [Understand] [CO: 2]
19. Explain the difference between white matter and grey matter in the central nervous system. [Understand] [CO: 4]
20. How does the cerebellum contribute to the learning of motor skills? [Understand] [CO: 3]
21. Comment on cerebral dominance. [Understand] [CO: 4]
22. Evaluate the role of the hypothalamus in maintaining homeostasis. [Evaluate] [CO: 4]

[2 x 10 = 20]

Part **C**
Short Essay Type Questions
Answer **5** Questions
Each question carries **4** marks

23. Describe the disadvantage of using human in experimental process. [Understand] [CO: 1]
24. Describe the structure of a neuron with labelled diagram. [Understand] [CO: 2]
25. Explain the significance of the threshold potential in the initiation of an action potential. [Understand] [CO: 2]
26. Explain the process of synaptic transmission and how it leads to the generation of excitatory or inhibitory postsynaptic potentials. [Understand] [CO: 2]
27. How does the forebrain contribute to higher cognitive functions such as thinking and decision-making? [Analyse] [CO: 3]
28. Explain the role of the cerebral cortex in higher cognitive functions such as thinking, memory, and decision-making. [Understand] [CO: 4]
29. Describe the impact of split-brain studies on our understanding of brain lateralization. [Understand] [CO: 3]

[4 x5 = 20]